Almost-Kähler Einstein Four-Manifolds: Constructions and Rigidity

Shuxing Yang, Junyao Wu, Rui Zhao,
Hongsheng Chen*, Yihao Yang*

College of Information Science and Electronic Engineering,
Zhejiang University
Qiushi Engine Team

*Corresponding authors:
Hongsheng Chen (hansomchen@zju.edu.cn)
Yihao Yang (yangyihao@zju.edu.cn)

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Abstract

We study the interaction between the Einstein equation, a compatible symplectic form, and the self-dual curvature operator in dimension four. An explicit complete noncompact strictly almost-Kähler metric with negative Einstein constant illustrates the distinction between local integrability and the closed problem. On closed manifolds, torsion zeros lead to a Green-potential criterion for integrability: a nonpositive potential of the determinant of the self-dual curvature block at one torsion zero forces the structure to be Kähler. Further formulations use canonical-class orthogonality, products of harmonic forms, the normalized Einstein action, and pseudoholomorphic representatives of the canonical class. We prove an energy-transfer statement for noncollapsed Einstein degeneration and describe the second-order obstruction to coupled almost-Kähler deformations of a negative Kähler–Einstein structure. The local Darboux system and its compatibility conditions are separated from these global arguments. The general closed negative Goldberg problem remains open; the results identify geometric conditions under which it has an affirmative answer and explain which degeneration mechanisms are not covered.

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1 The geometric problem and curvature conventions

An almost-Hermitian four-manifold consists of a Riemannian metric g and an orthogonal almost-complex structure J. Its fundamental form is

$$\Omega(X,Y) = g(JX,Y).$$

The structure is almost-Kähler if $d\Omega = 0$, and is Kähler if $\nabla\Omega = 0$. We use the orientation determined by $\Omega \wedge \Omega$. Thus

$$*\Omega = \Omega, \qquad |\Omega|^2 = 2, \qquad \Omega \wedge \Omega = 2\, dV_g. \tag{1}$$

A constant-length self-dual harmonic two-form satisfying (1) conversely determines an almost-Kähler structure.

Throughout the closed negative Einstein discussion,

$$\mathrm{Ric} = -ag, \qquad a > 0, \qquad s = -4a. \tag{2}$$

The scalar Laplacian is $\Delta = -\operatorname{div}\nabla$, so that it is nonnegative. Norms of $W^{\pm}$ are Hilbert–Schmidt norms of endomorphisms of $\Lambda^{\pm}$. The self-dual curvature block is

$$\mathcal{R}_+ = W^+ + \frac{s}{12}\,\mathrm{Id} = W^+ - \frac{a}{3}\,\mathrm{Id}\,.$$

We write $V = \mathrm{Vol}_g(M)$, $\chi = \chi(M)$, $\tau = \tau(M)$, and $T_+ = 2\chi + 3\tau$. The relevant Chern–Weil identities for an Einstein metric are

$$8\pi^2\chi = \int_M \left(|W^+|^2 + |W^-|^2 + \frac{s^2}{24} \right) dV_g, \tag{3}$$

$$12\pi^2\tau = \int_M \left(|W^+|^2 - |W^-|^2 \right) dV_g, \tag{4}$$

$$4\pi^2 T_+ = \int_M \left(2|W^+|^2 + \frac{s^2}{24} \right) dV_g. \tag{5}$$

Set $q = |\nabla\Omega|^2$, $n = \Omega/\sqrt{2}$, and $Z = \{q = 0\}$. Relative to $\Lambda^+ = \mathbb{R}n \oplus n^\perp$, write

$$W^+ = \begin{pmatrix} t & v^* \\ v & -t\,\mathrm{Id}\,/2 + B \end{pmatrix}, \qquad \operatorname{tr} B = 0, \qquad t = \frac{s}{6} + \frac{q}{4}, \qquad \eta = W^+\Omega - t\Omega = \sqrt{2}\,v. \tag{6}$$

Here $n^\perp = \Lambda^-_J$, the real bundle of J-anti-invariant two-forms. The identity for t follows by pairing the harmonic-form Weitzenböck equation with Ω.

The closed Goldberg question asks whether every closed Einstein almost-Kähler manifold is Kähler. Sekigawa's theorem settles the nonnegative scalar-curvature case [1]. We address the negative four-dimensional case. A complete noncompact example is compatible with this question; conversely, local calculations alone cannot decide a theorem whose essential hypothesis is closedness.

The local curvature calculations used below are given in [16, 17, 18, 19]. The global arguments concern canonical currents, Einstein deformations, and noncollapsed orbifold limits under the hypotheses stated in the corresponding sections.

2 A complete strictly almost-Kähler negative Einstein metric

2.1 The formula

Let $\sigma_1, \sigma_2, \sigma_3$ be a left-invariant coframe on $\mathrm{SL}(2,\mathbb{R})$ satisfying

$$d\sigma_1 = -2\sigma_2 \wedge \sigma_3, \qquad d\sigma_2 = 2\sigma_3 \wedge \sigma_1, \qquad d\sigma_3 = 2\sigma_1 \wedge \sigma_2. \tag{7}$$

For $1/3 < x < 1$, put $D = 1 + 3x^2$ and

$$\begin{aligned} G &= \frac{4D}{(1-x^2)^2(9x^2-1)}, \\ A_1 &= \frac{4(3x-1)D}{3(1-x)(1+x)^2}, \qquad A_2 = \frac{4(3x+1)D}{3(1-x)^2(1+x)}, \qquad A_3 = \frac{64x^2}{(1-x^2)D}, \\ X_1 &= \frac{(1-x)^3(3x+1)}{16x^3}, \qquad X_2 = \frac{(1+x)^3(3x-1)}{16x^3}. \end{aligned}$$

Define

$$g = G\,dx^2 + A_1\sigma_1^2 + A_2\sigma_2^2 + A_3\sigma_3^2. \tag{8}$$

With $e^0 = \sqrt{G}\,dx$, $e^i = \sqrt{A_i}\sigma_i$, let

$$\omega_i = e^0 \wedge e^i + e^j \wedge e^k \quad ((i,j,k) \text{ cyclic}), \qquad \Omega = \sqrt{X_1}\,\omega_1 + \sqrt{X_2}\,\omega_2.$$

Since $X_1 + X_2 = 1$, this form has norm $\sqrt{2}$.

Theorem 2.1. *The pair* (8), Ω *extends smoothly to the total space of*

$$\mathrm{SL}(2,\mathbb{R}) \times_{\mathrm{SO}(2)} \mathbb{R}^2 \longrightarrow \mathbb{H}^2,$$

which is diffeomorphic to $\mathbb{R}^4$. *The extended metric is complete and satisfies*

$$\mathrm{Ric} = -3g, \qquad W^- = 0, \qquad d\Omega = 0, \qquad |\nabla\Omega|^2 = \frac{36(1-x^2)^3}{(1+3x^2)^3} > 0.$$

Proof. Here are the differential and extension calculations. Their full curvature expansion is recorded in [17], together with an independent Cartan calculation. Let $dr = \sqrt{G}\,dx$ and $h_i = (\log\sqrt{A_i})'$. The structure equations take the form

$$de^i = h_i e^0 \wedge e^i + 2w_i e^j \wedge e^k.$$

The signs of w_i are determined by (7). Consequently,

$$d(f_i\omega_i) = (f_i' + (h_j + h_k - 2w_i)f_i)e^0 \wedge e^j \wedge e^k.$$

Substitution of $f_1 = \sqrt{X_1}$, $f_2 = \sqrt{X_2}$, $f_3 = 0$ makes each coefficient zero. The Levi-Civita connection obtained from these equations has zero mixed curvature block, $\mathcal{R}_- = -\mathrm{Id}$, and $\mathcal{R}_+ = \mathrm{diag}(r_1, r_2, r_3)$, where

$$\begin{aligned} r_1 &= -\frac{3(x-1)^2(21x^4 + 18x^3 + 12x^2 - 2x - 1)}{D^3}, \\ r_2 &= -\frac{3(x+1)^2(21x^4 - 18x^3 + 12x^2 + 2x - 1)}{D^3}, \\ r_3 &= \frac{9(x-1)^2(x+1)^2(5x^2-1)}{D^3}. \end{aligned}$$

Their sum is -3. These equalities give $\mathrm{Ric} = -3g$ and $W^- = 0$. The same connection gives the displayed value of q.

Choose $r = 0$ at $x = 1/3$. Taylor expansion yields

$$\begin{aligned} x - \tfrac{1}{3} &= \tfrac{2}{9}r^2 + O(r^4), & A_1 &= r^2 + O(r^4), \\ A_2 = A_3 + O(r^4) &= 6 + 7r^2 + O(r^4), & A_2 - A_3 &= \tfrac{10}{3}r^4 + O(r^6), \\ X_2 &= \tfrac{8}{3}r^2 + O(r^4), & q(0) &= \tfrac{32}{3}. \end{aligned}$$

Take the collapsing circle to have period 2π. In a local normal coordinate $z = re^{i\psi}$, the normal metric has the smooth expression $|dz + iz\beta|^2$ to leading order. The horizontal frame has weight two under this circle; its anisotropic weight-four coefficient vanishes to order r^4. The mixed coefficients of Ω have the forms $zF(r^2)$ and $z^3H(r^2)$, with smooth F, H. These expressions give the smooth extension across the zero section.

As $x \to 1$,

$$G(x) \sim \frac{1}{2(1-x)^2},$$

so the radial distance is infinite. On every bounded radial interval, the extended metric dominates $dr^2 + c_R\pi^* g_{\mathbb{H}^2}$ for some $c_R > 0$. A finite-length escaping curve therefore cannot have unbounded r or escape in the complete hyperbolic base. The circle fibers and the normal disk over a bounded base set are compact. Thus every finite-length curve remains in a compact set, proving completeness. □

The example is nonhomogeneous: $|W^+|^2$ equals $2/3$ on the zero section and tends to 6 at infinity. Moreover,

$$|\,\mathrm{pr}_{n^\perp}(W^+ n)|^2 = \frac{9(1-x^2)^3(9x^2-1)}{(1+3x^2)^4}. \tag{9}$$

Thus the compatible form is not a Weyl eigenform on the regular region. Pulling the pair back to the additive Lie group $(\mathbb{R}^4, +)$ gives a complete noninvariant, non-Ricci-flat strictly almost-Kähler Einstein structure, of the type asked about after Corollary 2.1 of Fino [5]. This is a concrete affirmative example for that existence question. The ansatz of Armstrong [4] treats related local geometry; equation (9) shows that this example lies outside its weakly star-Einstein subclass. No historical first-solution assertion is needed here. Its infinite volume and noncompactness distinguish it from the closed problem.

2.2 Two restrictions on symmetric completions

The following local classification records a complementary result in a more symmetric class.

Theorem 2.2 (Invariant nut classification)**.** *Suppose a smooth four-dimensional almost-Kähler Einstein pair has* $\mathrm{Ric} = -3g$*, a smooth nut, regular three-sphere orbits, and a full* $\mathrm{SU}(2)$ *action preserving both* g *and* Ω*. In the chirality admitting a nonzero invariant self-dual form at the nut, the pair is locally complex hyperbolic and Kähler.*

The proof in [16] uses the momentum constraint to diagonalize the invariant metric in a constant frame. If $a_i = t f_i(t^2)$ and

$$f_i(u) = 1 + (\tfrac{1}{6} + d_i)u + \cdots, \qquad \sum_i d_i = 0,$$

the closed-form and unit-length equations force, up to permutation,

$$(d_1, d_2, d_3) = (\tfrac{1}{6}, -\tfrac{1}{12}, -\tfrac{1}{12}), \qquad (X_1(0), X_2(0), X_3(0)) = (1, 0, 0).$$

The higher recurrence has eigenvalues $4(n+2)(n-1)$ and $2(n+2)(2n+1)$ for $n \geq 2$, hence is unique. The resulting solution is

$$a_1 = \frac{\sinh(\sqrt{2}t)}{\sqrt{2}}, \qquad a_2 = a_3 = \sqrt{2}\sinh(t/\sqrt{2}), \qquad \Omega = \pm\omega_1.$$

The algebraic computation is given in Computation. The theorem concerns symmetry of the pair, not merely symmetry of its metric.

A different obstruction concerns closing hyperbolic bolts. If a closed manifold is assembled from two-disk bundles over surfaces Σ_i and product collars, additivity of Euler characteristic gives $\chi(M) = \sum_i \chi(\Sigma_i)$. When all bolts have genus greater than one, this is negative, contradicting (3). An orbit space that is a circle and has no bolts gives $\chi = 0$, which forces a closed Einstein metric to be flat. These topological observations rule out these particular completions of noncompact models.

3 Torsion zeros and their curvature constraints

Choose a local frame $\phi, J\phi$ for Λ^-_J, each of norm $\sqrt{2}$. The torsion is expressed by a real one-form ϑ:

$$\nabla\Omega = \vartheta \otimes \phi - J\vartheta \otimes J\phi, \qquad q = 4|\vartheta|^2, \qquad J\vartheta(X) = -\vartheta(JX). \tag{10}$$

Changing the unitary frame changes ϑ equivariantly; its zero set is intrinsic.

Theorem 3.1 (The torsion-zero equations)**.** *For an Einstein almost-Kähler four-manifold, at a point of* Z *the real-linear map* $L = (\nabla\vartheta)^\sharp$ *has a unitary-coordinate expression*

$$L(z) = cz + Q\bar{z}, \qquad \operatorname{tr}(Q\bar{Q}) = -8|c|^2, \qquad \det_{\mathbb{R}} L = 9|c|^4 + |\det_{\mathbb{C}} Q|^2 \geq 0. \tag{11}$$

The bar on Q *denotes entrywise conjugation. A nontransverse zero has rank zero or two. At every torsion zero,*

$$|\nabla^2\Omega|^2 = 4|B|^2 - 6|\eta|^2. \tag{12}$$

These are the local curvature and differentiated closedness equations proved in [18].

Proposition 3.2. *Every closed negative Einstein almost-Kähler four-manifold has a torsion zero. If all torsion zeros are transverse, their number is* $5\chi + 6\tau$*, and each has index* $+1$*.*

Proof. The Nijenhuis tensor is a section of the complex rank-two bundle

$$\mathcal{N} = T^{1,0}M \otimes \bar{K},$$

where K is the canonical bundle. If x_1, x_2 are the Chern roots of $T^{1,0}M$, those of $\mathcal{N}$ are $2x_1 + x_2, x_1 + 2x_2$. Consequently,

$$c_2(\mathcal{N})[M] = 2c_1^2 + c_2 = 5\chi + 6\tau.$$

Using (3)–(4),

$$5\chi + 6\tau = \frac{1}{8\pi^2}\int_M \left(9|W^+|^2 + |W^-|^2 + \frac{5s^2}{24}\right) dV_g > 0.$$

A nowhere-zero section would give a trivial complex line subbundle and make the top Chern number zero. Thus $Z \neq \varnothing$, as in Armstrong's argument [3]. At a transverse zero (11) makes the real determinant positive. The index formula gives the count. □

Corollary 3.3. *At a torsion zero,*

$$\lambda_{\min}(W^+) \leq \frac{s}{6}, \qquad |W^+|^2 \geq \frac{s^2}{24}.$$

A closed negative Einstein metric with $\lambda_{\min}(W^+) > s/6$ everywhere admits no compatible almost-Kähler form in that orientation. In particular this holds when $W^+ = 0$.

Proof. At a zero, $\langle W^+ n, n\rangle = s/6$. The eigenvalue bound is the Rayleigh inequality. Since W^+ is trace free, the least possible squared norm of a symmetric matrix with this prescribed diagonal entry is $3(s/6)^2/2 = s^2/24$. □

These pointwise bounds do not give a lower bound for the volume of the corresponding curvature region.

3.1 Zero curves and leading jets

The zero-set analysis in [19] yields further necessary conditions. There is no compact boundaryless embedded component $C \subset Z$ with $\ker DN_J = TC$ everywhere. More generally, if C is a closed J-holomorphic zero curve and rank-two points are dense, it is totally geodesic. The normal derivative of the torsion is a section

$$\xi \in \mathrm{Hom}_{\mathbb{C}}(\bar{\nu}_C, T^{1,0}C \otimes \bar{K}|_C) \cong \bar{K}^2|_C,$$

whose degree in the stated convention is $s\,\mathrm{Area}(C)/(4\pi) < 0$. Thus it must vanish somewhere. The differentiated equations have the form

$$D^{1,0}\xi = r, \qquad D^{1,0}r = 0,$$

where r is a normal derivative of transverse curvature. Their inhomogeneous first equation does not imply that one torsion zero propagates to all of M.

If a torsion zero has finite order $m \geq 2$, the first nonzero homogeneous term necessarily has, in unitary coordinates, the form

$$f_m(z) = \kappa(\bar{z}_2)^m e_1. \tag{13}$$

One obtains this by writing $f_m = zh(\bar{z}) + g(\bar{z})$, with degrees $m-1, m$, and imposing the first differentiated curvature compatibility. Its distinct bidegrees force $h = 0$. The coefficient matrices of Dg span an isotropic subspace for the Hermitian form $\mathrm{tr}(\bar{U}V)$ of signature $(3,1)$, so this span has dimension at most one. Integrability gives a rank-one matrix $v \otimes \ell$ with $\ell \cdot \bar{v} = 0$, which is (13). This is a necessary leading-jet classification, not an existence theorem for its nonlinear extension.

4 Darboux coordinates and the linear compatibility complex

Fix the standard symplectic form on $\mathbb{C}^2$. Symmetric tensors satisfying $HJ + JH = 0$ describe infinitesimal compatible metrics. The nonlinear parameterization

$$G = e^H, \qquad H^T = H, \qquad HJ + JH = 0 \tag{14}$$

ensures $G > 0$, $\det G = 1$, and $GJG = J$. It removes these algebraic constraints from the Einstein equations.

For a nonzero real covector ξ, use the Ricci principal symbol

$$S_\xi(H) = \frac{1}{2}\left(\xi \otimes H\xi + H\xi \otimes \xi - |\xi|^2 H\right). \tag{15}$$

The sign of the Fourier convention has no effect on its kernel and image.

Proposition 4.1. *On the six-dimensional space of anti-J symmetric tensors,*

$$\begin{aligned}
&\ker S_\xi = \mathbb{R}(\xi \otimes J\xi + J\xi \otimes \xi),\\
&\operatorname{im} S_\xi = \left\{K : K\xi - \tfrac{1}{2}(\operatorname{tr} K)\xi = 0, \quad K(J\xi, J\xi) - \tfrac{1}{2}|\xi|^2 \operatorname{tr} K = 0\right\}.
\end{aligned}$$

Moreover,

$$|S_\xi(H)|^2 + \frac{1}{4}|H(\xi, J\xi)|^2 \geq \frac{1}{8}|\xi|^4 |H|^2. \tag{16}$$

Proof. By unitary invariance and homogeneity take $\xi = e_1$, $Je_1 = e_2$, $Je_3 = e_4$. Then

$$H = \begin{pmatrix} a & b & c & d \\ b & -a & d & -c \\ c & d & e & f \\ d & -c & f & -e \end{pmatrix}, \qquad S_{e_1} H = \frac{1}{2}\begin{pmatrix} a & 0 & 0 & 0 \\ 0 & a & -d & c \\ 0 & -d & -e & -f \\ 0 & c & -f & e \end{pmatrix}.$$

The kernel is the b direction, the five displayed image conditions are independent, and direct comparison of the squared norms gives (16). Setting $b = 0$ gives an explicit right inverse on the image. The kernel is the principal Hamiltonian coordinate-change direction. □

For the polynomial prolongation, let $\mathcal{P}_n$ denote homogeneous real polynomials of degree n in four variables and $\Delta_0 = -\sum \partial_i^2$. Define

$$\begin{aligned}
(L_n h)_{ij} &= \frac{1}{2}(\partial_k \partial_i h_{kj} + \partial_k \partial_j h_{ki} + \Delta_0 h_{ij}),\\
\mathscr{B}(E)_j &= \partial_i E_{ij} - \tfrac{1}{2}\partial_j \operatorname{tr} E,\\
\mathscr{P}(E) &= J_{ia} J_{jb} \partial_a \partial_b E_{ij} + \tfrac{1}{2}\Delta_0 \operatorname{tr} E.
\end{aligned}$$

Theorem 4.2 (Homogeneous polynomial exactness)**.** *For every $n \geq 0$,*

$$\operatorname{im} L_n = \ker(\mathscr{B}, \mathscr{P}), \qquad \dim \operatorname{coker} L_n = 4\binom{n+2}{3} + \binom{n+1}{3}. \tag{17}$$

Negative-degree polynomial spaces and their binomial coefficients are zero. The exact calculation and polynomial lifts are recorded in [16], section "Darboux representation and linear compatibility."

For clarity, the dimension calculation uses the torsion map on $\ker L_n$. In complex notation its image is

$$\{z h_0(\bar{z}) + g_0(\bar{z}) : \deg h_0 = n,\ \deg g_0 = n+1\},$$

of real dimension $6n+10$. For $h_0 = w_1^n$, a lift is

$$\nu_{11} = -w_1^n w_2 z_1 - \frac{n}{2} w_1^{n-1} w_2^2 z_2, \quad \nu_{12} = -w_1^n w_2 z_2, \quad \nu_{22} = 0;$$

the second summand is zero for $n = 0$. Unitary transformations generate all h_0, and the polynomial Poincaré lemma generates g_0. The resulting kernel dimension is

$$\dim \ker L_n = \binom{n+7}{3} + (n+5)^2 + 2n - 10.$$

The Fischer adjoint of $(\mathscr{B}, \mathscr{P})$ is multiplication by its symbol. Proposition 4.1 gives rank five at each nonzero real covector; hence this adjoint has zero polynomial kernel. The dimension count proves (17).

Even after Hamiltonian gauge is removed, the torsion-free part of the kernel has dimension $(n+1)(n+5)$. Its five dimensions at $n = 0$ represent anti-self-dual curvature freedom. At each nonlinear order the previously chosen coefficients determine the source; (17) tests whether that source can be lifted. It neither proves that every nonlinear source passes the test nor gives convergence of an infinite formal series.

5 A determinant potential and integrability

Set

$$A = -\mathcal{R}_+, \qquad U = \sigma_2(A) = \frac{a^2}{3} - \frac{1}{2}|W^+|^2. \tag{18}$$

The classical Einstein Weyl equation [2] is

$$\nabla^* \nabla W^+ + \frac{s}{2} W^+ - 6(W^+)^2 + 2|W^+|^2 \,\mathrm{Id} = 0. \tag{19}$$

Lemma 5.1. *On an Einstein four-manifold satisfying* (2)*,*

$$(\Delta + 2a)U = |\nabla A|^2 - 18 \det \mathcal{R}_+. \tag{20}$$

Proof. Pair (19) with W^+, use $\operatorname{tr} W^+ = 0$ and $\operatorname{tr}((W^+)^3) = 3 \det W^+$, and apply

$$\tfrac{1}{2}\Delta |W^+|^2 = \langle \nabla^* \nabla W^+, W^+ \rangle - |\nabla W^+|^2.$$

For a trace-free three-by-three matrix W,

$$\det(W - a \,\mathrm{Id}\,/3) = \det W + \frac{a}{6}|W|^2 - \frac{a^3}{27}.$$

Substitution in $\Delta U = -\frac{1}{2}\Delta |W^+|^2$ gives (20). □

On a closed connected manifold, $L_a = \Delta + 2a$ has a strictly positive Green kernel G_a. Define

$$\Psi = L_a^{-1}(\det \mathcal{R}_+), \qquad \mathcal{D} = L_a^{-1}(|\nabla A|^2) \geq 0.$$

Then $U = \mathcal{D} - 18\Psi$.

Theorem 5.2 (Green-potential rigidity)**.** *Let (M, g, J, Ω) be closed, connected, and almost-Kähler with* $\mathrm{Ric} = -ag$*, $a > 0$. At every $p \in Z$,*

$$18\Psi(p) = \mathcal{D}(p) + \frac{1}{8}|\nabla^2 \Omega(p)|^2 + \frac{5}{4}|\eta(p)|^2. \tag{21}$$

If $\Psi(p) \leq 0$ at one torsion zero, then $\nabla \Omega = 0$. In particular, the same conclusion holds under the metric condition $\det \mathcal{R}_+ \leq 0$ *everywhere.*

Proof. At $p \in Z$, (6) gives $t = -2a/3$, and hence

$$U(p) = -\tfrac{1}{2}(|\eta(p)|^2 + |B(p)|^2).$$

Using (12) gives

$$U(p) = -\tfrac{1}{8}|\nabla^2\Omega(p)|^2 - \tfrac{5}{4}|\eta(p)|^2,$$

which proves (21). If its left side is nonpositive, every term on the right vanishes. Strict positivity of G_a then forces $\nabla A \equiv 0$. At p, $\eta = B = 0$, so $|W^+|^2 = 2a^2/3 = s^2/24$. Parallel curvature makes this equality global.

The integrated almost-Kähler Einstein identity stated in (26) below now has zero left side, hence $q \equiv 0$. Equivalently $A = a, n \otimes n$ and Ω is parallel. For the final assertion, positivity preservation of L_a^{-1} makes $\Psi \leq 0$, and Proposition 3.2 supplies a torsion zero. □

The proof is the combination of a scalar elliptic equation and the local zero identity, rather than a pointwise sign conclusion about all Einstein metrics. For a strictly almost-Kähler closed candidate it requires

$$\Psi(p) > 0 \quad (p \in Z), \qquad \det \mathcal{R}_+ > 0 \text{ on some open set.}$$

Let $\Sigma = \{\lambda_1(W^+) = \lambda_2(W^+)\}$, where eigenvalues are ordered decreasingly. The local equations also give

$$Z \cap \Sigma = \{\nabla\Omega = 0,\ \nabla^2\Omega = 0\}.$$

At such a point $\eta = 0$, and (21) reduces to

$$18\Psi(p) = \int_M G_a(p, y)|\nabla A(y)|^2 \, dV_y.$$

Thus even a degenerate zero measures the global variation of curvature through the Green kernel. The detailed original proof is [20].

6 Canonical orthogonality and harmonic products

6.1 A global orthogonality relation

The first canonical Hermitian connection has real curvature form γ with $[\gamma] = 2\pi c_1(M, J)$. For an Einstein almost-Kähler surface its decomposition gives

$$\beta = \gamma - \frac{s}{4}\Omega, \qquad d\beta = 0, \qquad \beta^{J,-} = \eta. \tag{22}$$

Lemma 6.1. *If α is a closed real J-anti-invariant two-form on a closed almost-complex four-manifold, then $K \cdot [\alpha] = 0$.*

Proof. If α is nowhere zero, it trivializes the complex line underlying Λ_J^-, so the real canonical class vanishes. Otherwise the canonical-divisor theorem of Bonthrone–Zhang [8] represents K by a J-holomorphic one-subvariety supported on the zero set of α. Evaluation of α on this current is zero. The assertion is also immediate when $\alpha \equiv 0$. □

Proposition 6.2. *For the closed Einstein almost-Kähler data above,*

$$\int_M \langle \eta, \alpha \rangle \, dV_g = 0 \quad \textit{for every closed real } \alpha \in \Gamma(\Lambda_J^-).$$

If a smooth positive function f satisfies $d(f\eta) = 0$, then J is integrable.

Proof. Pointwise $\Omega \wedge \alpha = 0$ and $\beta \wedge \alpha = \langle \eta, \alpha \rangle dV_g$. Thus (22) and Lemma 6.1 give the integral. Taking $\alpha = f\eta$ makes $\int f|\eta|^2 = 0$. Hence $\eta = 0$. For an Einstein metric this is the third Gray curvature condition; the compact integrability theorem of Apostolov–Armstrong–Drăghici [6] gives $\nabla\Omega = 0$. □

On $\{\eta \neq 0\}$ write $d\eta = \vartheta_\eta \wedge \eta$. An integrating factor requires $d \log f = -\vartheta_\eta$. Thus closedness of ϑ_η is a local requirement, and its periods and extension across zeros are global requirements. The complete metric in Section 2 has $d\vartheta_\eta \neq 0$ on an open set [21]; such a factor is not an automatic consequence of the local Einstein equations.

6.2 A finite-dimensional formulation for a fixed metric

For a fixed closed oriented metric, choose an average-orthonormal basis $\psi_1, \dots, \psi_m$ of $\mathcal{H}^+_g$, where $m = b^+_2(M)$. Put

$$G_{ij}(x) = \langle \psi_i(x), \psi_j(x) \rangle, \qquad B_g(x) = G(x) - I,$$
$$H_{ij} = \fint_M \langle \nabla\psi_i, \nabla\psi_j \rangle \, dV_g, \qquad \mathsf{C}_g(X) = \fint_M \langle B_g(x), X \rangle B_g(x) \, dV_g.$$

Proposition 6.3 (Harmonic product criterion)**.** *Compatible almost-Kähler forms for g, modulo sign, correspond exactly to matrices*

$$X \in \ker \mathsf{C}_g, \qquad X \succeq 0, \qquad \operatorname{tr} X = 1, \qquad \operatorname{rank} X = 1. \tag{23}$$

Under this correspondence,

$$\fint_M |\nabla\Omega|^2 \, dV_g = 2 \operatorname{tr}(HX).$$

Proof. Write $X = uu^T$, $|u| = 1$, and set $\Omega = \sqrt{2} \sum u_i \psi_i$. It is closed and self-dual. Furthermore,

$$\langle \mathsf{C}_g(X), X \rangle = \fint_M \langle B_g(x), X \rangle^2 \, dV_g.$$

Its vanishing is precisely the constant-length identity $|\Omega|^2 = 2$. Every compatible form is harmonic and has such an expansion. The derivative identity follows by bilinearity. □

The positivity and rank-one requirement in (23) are essential: an arbitrary matrix in the linear kernel need not define a form. For a negative Einstein metric admitting a parallel self-dual form, the uniqueness theorem of Di Scala–Nagy [7] makes every compatible form in that orientation parallel. The remaining fixed-metric question is whether the rank-one positive part of $\ker \mathsf{C}_g$ can meet directions with $\operatorname{tr}(HX) > 0$.

6.3 Length defect on a negative Kähler–Einstein background

Let (V, g, J, ω) be a compact negative Kähler–Einstein surface or orbifold. Hodge theory gives

$$\mathcal{H}^+_g = \mathbb{R}\omega \oplus \operatorname{Re} H^{2,0}(V), \qquad |b\omega + \beta|^2 = 2b^2 + |\beta|^2.$$

A nonzero holomorphic canonical form of constant norm would trivialize the canonical orbifold line bundle. This contradicts $c_1(V) = [\rho/(2\pi)] \neq 0$. Therefore $Q_0(\beta) = |\beta|^2 - \fint |\beta|^2$ is nonzero whenever $0 \neq \beta \in \operatorname{Re} H^{2,0}(V)$. Compactness of the unit sphere of this finite-dimensional space proves

$$\|Q_0(\beta)\|_{L^2} \geq c_V \|\beta\|^2_{L^2} \tag{24}$$

for some $c_V > 0$. This is quadratic coercivity of the length defect. Its squared variance is quartic in β, whereas the derivative energy is quadratic; the two cannot be compared by a uniform linear bound near the Kähler direction. The constant here depends on the fixed background.

7 Einstein action, torsion energy, and connected components

Define

$$\widehat{S}(g) = V^{-1/2} \int_M s \, dV_g, \qquad D_W(g) = \int_M \left(\frac{s^2}{24} - |W^+|^2 \right) dV_g.$$

For an Einstein metric, (5) gives

$$D_W = \frac{\widehat{S}^2}{16} - 2\pi^2 T_+. \tag{25}$$

Theorem 7.1 (Integrated almost-Kähler Einstein identity)**.** *For a closed Einstein almost-Kähler four-manifold in the conventions above, the integrated curvature identity is*

$$D_W = \mathcal{T}, \qquad \mathcal{T} = \int_M \left(|B|^2 + \frac{3q^2}{32} \right) dV_g \geq 0. \tag{26}$$

It is the Einstein specialization of the Sekigawa identities, with the block conventions (6)*; the derivation is given in [22, 16]. In particular* $\mathcal{T} = 0$ *if and only if* Ω *is parallel.*

Combining (25) and (26) yields

$$\widehat{S}^2 = 32\pi^2 T_+ + 16\mathcal{T}. \tag{27}$$

This compares topology with a nonnegative torsion integral.

Proposition 7.2. *The normalized Einstein action* $\widehat{S}$ *is locally constant on the moduli space of Einstein metrics modulo orientation-preserving diffeomorphisms and scale. It is constant on every connected component and assumes only finitely many values on each compact subset.*

Proof. Einstein metrics are critical points of $\widehat{S}$. Koiso's local deformation theorem gives a finite-dimensional real-analytic Kuranishi model in a slice for the diffeomorphism action [12, 13]. In a sufficiently small neighborhood, points in the local analytic set can be joined to the base point by finitely many differentiable subanalytic arcs. Along each arc the derivative of $\widehat{S}$ vanishes, because every metric on it is critical. Hence the action is locally constant. It descends to the quotient because it is invariant under diffeomorphism and scale. A locally constant function is constant on connected components, and a finite subcover of a compact subset proves the final assertion. No integrability assumption on all infinitesimal Einstein deformations is used. □

Corollary 7.3. *If a connected Einstein component contains a negative Kähler–Einstein metric in a fixed orientation, every almost-Kähler structure carried by a metric in that component in the same orientation is Kähler.*

Proof. At the Kähler–Einstein metric, $|W^+|^2 = s^2/24$ and $D_W = 0$. Equations (25) and Proposition 7.2 make $D_W = 0$ throughout the component. Apply (26). □

For the opposite orientation, (4) gives instead $D_W^- = D_W^+ + 12\pi^2\tau$. Thus the corresponding torsion integral is $12\pi^2\tau$. A negative signature excludes an oppositely oriented compatible form; zero signature forces it to be parallel. Positive signature does not yield a contradiction.

7.1 Opposite orientations on a Kähler background

If I is Kähler and J induces the opposite orientation, then I, J commute. The symmetric involution $-IJ$ splits $TM = D_+ \oplus D_-$ into orthogonal I-complex lines. Writing

$$\omega_I = \omega_+ + \omega_-, \qquad \Omega_J = \omega_+ - \omega_-,$$

closedness gives $d\omega_\pm = 0$. Their kernels are integrable, with minimal complex leaves. Both foliations are totally geodesic exactly when J is parallel. They are smooth complex distributions; holomorphicity of the subbundles is additional information.

For $\mathrm{Ric} = Eg$, constant rescaling

$$g_{a,b} = a, g|_{D_+} + b, g|_{D_-}, \qquad a, b > 0,$$

remains I-Kähler, has the same Ricci form, and has Ricci eigenvalues $E/a, E/b$. This is the relation with constant Ricci-eigenvalue geometry studied in [11]. If $x = c_1(D_+^{1,0})$ and $y = c_1(D_-^{1,0})$, then $xy = \chi$ and $x^2 + y^2 = 3\tau$; the cross pairing of the two canonical classes involves $x^2 - y^2$, which is not determined by those two characteristic numbers alone.

8 Pseudoholomorphic canonical curves and integrated inequalities

The first canonical curvature has, on an oriented unit J-complex plane Π, the formula

$$\gamma(\Pi) = \frac{s}{4} + |\vartheta_{\Pi^\perp}|^2. \tag{28}$$

Thus $\beta = \gamma - s\Omega/4$ is closed and J-semipositive, and $\Omega + t\beta$ tames J for $t \geq 0$. Let $K = -c_1(M, J)$.

Proposition 8.1. *Suppose a J-holomorphic current C_K represents* $\mathrm{PD}(K)$. *Then*

$$\mathcal{T} - \int_M |v|^2 \, dV_g = 2\pi \int_{C_K} |\vartheta_{\Pi^\perp}|^2 \, dA \geq 0. \tag{29}$$

Proof. Write $Q_\vartheta = \int q \, dV_g$. The same curvature identities that yield (26) give

$$\frac{|s|}{16} Q_\vartheta = \mathcal{T} + \int |v|^2 \, dV_g, \qquad 2\pi c_1 \cdot [\Omega] = \frac{sV}{2} + \frac{Q_\vartheta}{4}. \tag{30}$$

Cohomology and $c_1^2 = T_+$ therefore give

$$\begin{aligned} 2\pi[\beta] \cdot K &= -4\pi^2 c_1^2 + \frac{s^2 V}{8} + \frac{s Q_\vartheta}{16} \\ &= \mathcal{T} - \int |v|^2 \, dV_g. \end{aligned}$$

Evaluating (28) on the current proves the formula. Multiplicities are included in the current integral. □

Existence of the current is a substantive hypothesis. Taubes's theory establishes it under the appropriate $b_2^+ > 1$ or chamber hypotheses; the cohomology class alone does not imply existence. The use of pseudoholomorphic curves in Einstein obstructions has a classical precedent in LeBrun [9].

If $K = K_0 + \sum_{i=1}^r e_i$, $e_i \cdot e_j = -\delta_{ij}$, and $K_0 \cdot e_i = 0$, and if pseudoholomorphic currents represent K_0 and the exceptional classes, the same pairing yields

$$\mathcal{T} - \int |v|^2 \, dV_g \geq 4\pi^2 r + \frac{\pi|s|}{2} \sum_i \mathrm{Area}(E_i), \tag{31}$$

$$s^2 V \geq 32\pi^2 K_0^2 + 32\pi^2 r + 8\pi|s| \sum_i \mathrm{Area}(E_i). \tag{32}$$

For any J-holomorphic current C with $c_1 \cdot C = d > 0$, (28) and $q = 4|\vartheta|^2$ also give

$$\int_C q \, dA \geq 8\pi d + |s| \, \mathrm{Area}(C).$$

These are constraints on the curvature of the same metric and the geometry of its pseudoholomorphic curves.

A separate topological consequence of (26) is

$$8\pi^2(\chi - 3\tau) = \mathcal{T} + 3\int |W^-|^2\, dV_g. \tag{33}$$

If $b_2^- = 0$ and $b_2^+ \geq 1$, then $\chi - 3\tau = 2 - 2b_1 - 2b_2^+ \leq 0$. All terms in (33) vanish; the structure is Kähler. A closed strict candidate must therefore have $b_2^- \geq 1$ and $\chi > 3\tau$.

9 Noncollapsed degeneration and transfer of energy

Fix a closed oriented smooth four-manifold M. Consider Einstein metrics g_j satisfying

$$s_{g_j} = s_0 < 0, \qquad \mathrm{Vol}(M, g_j) \geq v_0 > 0, \qquad \mathrm{diam}(M, g_j) \leq D. \tag{34}$$

The upper volume bound follows from the Ricci lower bound and diameter bound. Equation (3) supplies the curvature integral bound. The Einstein orbifold compactness theorem [14] gives a subsequence converging smoothly on the regular set of a compact Einstein orbifold (V, g_∞), with finitely many singular points; volumes converge.

Write $m_+ \geq 0$ for the total lost W^+ energy, including every concentration scale and intervening neck. Then

$$D_W(V) = \lim_j D_W(g_j) + m_+. \tag{35}$$

Equivalently, if

$$T_+^{\mathrm{curv}}(V) = \frac{1}{4\pi^2}\int_V \left(2|W^+|^2 + \frac{s_0^2}{24}\right) dV,$$

then $T_+(M) = T_+^{\mathrm{curv}}(V) + m_+/(2\pi^2)$.

Theorem 9.1 (Almost-Kähler energy transfer)**.** *Assume* (34)*, and suppose each g_j has a compatible almost-Kähler form Ω_j inducing the fixed orientation. Then the limiting orbifold has a smooth compatible almost-Kähler form Ω_∞ and*

$$m_+ = 0, \qquad \mathcal{T}(g_j, \Omega_j) \longrightarrow \mathcal{T}(g_\infty, \Omega_\infty). \tag{36}$$

Every Ricci-flat ALE curvature limit is anti-self-dual in this orientation and has a parallel compatible Kähler form.

Proof. The harmonic forms Ω_j have constant norm. Elliptic estimates on each compact regular subset yield smooth convergence to a harmonic self-dual form of norm $\sqrt{2}$. Near an orbifold point, lift to a punctured ball. A cutoff at radius ε has $\|d\chi_\varepsilon\|_{L^1} = O(\varepsilon^3)$; boundedness of the form removes the point from the weak first-order equations. Elliptic regularity then gives a smooth invariant form on the ball. Thus the limit is an orbifold almost-Kähler structure.

The torsion density $|B|^2 + 3q^2/32$ is bounded by a constant times $|W^+|^2 + s_0^2$, by (6). Let $m_{\mathcal{T}} \geq 0$ be its lost integral. The integrated identity (26) also holds on the compact orbifold: it follows by invariant local charts and cutoff integration. Hence

$$\mathcal{T}(V) + m_{\mathcal{T}} = \lim_j \mathcal{T}(g_j) = \lim_j D_W(g_j) = D_W(V) - m_+ = \mathcal{T}(V) - m_+.$$

Nonnegativity forces both losses to vanish. At any blow-up scale the Einstein constant tends to zero. A nonzero W^+ curvature contribution would yield positive m_+, so every such ALE limit has $W^+ = 0$. The limiting compatible form satisfies the Ricci-flat harmonic-form equation $\nabla^*\nabla\Omega = 0$ and has constant norm, hence is parallel. □

Anti-self-dual curvature can still concentrate. In particular, the theorem does not eliminate Ricci-flat ALE bubbles or prove compactness without collapse.

Corollary 9.2. *Suppose a connected Einstein component has a noncollapsed same-orientation Kähler–Einstein orbifold limit. Every compatible almost-Kähler structure on that component is Kähler.*

Proof. At the endpoint $D_W(V) = 0$. The constant action on the component and (35) give $D_W = -m_+ \le 0$. At any almost-Kähler metric $D_W = \mathcal{T} \ge 0$. Thus both vanish. □

A sequence of strict almost-Kähler Einstein structures with $\mathcal{T}_j \to 0$ leaves every compact subset of the smooth Einstein moduli space. Under (34), its limit is a Kähler–Einstein orbifold and must have a singular point. Different connected components may be involved; local finiteness of components at such an orbifold endpoint is a separate issue.

For a general-type blowup $M = X\#k\overline{\mathbb{CP}}^2$, let $A = c_1^2(X) > 0$. The classical Ricci integral bound [10], when evaluated on an Einstein metric, gives

$$\widehat{S}^2 \ge 32\pi^2(A + k), \qquad D_W \ge 4\pi^2 k.$$

Together with (35) this excludes a same-orientation Kähler–Einstein orbifold main limit when $k > 0$. It is a restriction on that degeneration, not a nonexistence theorem for every Einstein metric on the blowup.

10 Coupled deformations of a negative Kähler–Einstein pair

10.1 The second-order length obstruction

Let (M, g, J, ω) be negative Kähler–Einstein with $\mathrm{Ric} = Eg$, $E < 0$. Consider second-order expansions

$$g^{-1}g_t = I + tH_1 + \tfrac{1}{2}t^2H_2 + O(t^3),$$
$$\omega_t = \omega + t\beta_1 + \tfrac{1}{2}t^2\beta_2 + O(t^3).$$

Use a joint diffeomorphism gauge for the metric and form. Assume H_1 is an anti-J Kodaira–Spencer direction and $\beta_1 \in \mathrm{Re}\, H^{2,0}$. The Einstein second-order normal form used here is

$$H_2^+ = H_1^2, \tag{37}$$

where the superscript means the J-invariant symmetric part, not Hodge self-duality. This is the specific second-order input of Nagy [15, Theorem 1.6]; its solvability hypotheses are required.

Transporting the closed self-dual form produces a closed correction β' with

$$(\beta')^- = -DP_g^-[H_1]\beta_1, \qquad \langle \omega, \beta' \rangle = 0.$$

Here P_g^- is the anti-self-dual projection, and the second equality fixes the Kähler component. The general remaining second-order choice is $\beta_2 = 2\beta' + \gamma$, $\gamma \in \mathcal{H}_g^+$. Expanding the compatibility length with (37) gives

$$|\beta_1|^2 + \langle \omega, \gamma \rangle = 0. \tag{38}$$

The second term is constant, since γ is self-dual harmonic on a Kähler surface. Consequently the mean-zero obstruction is precisely $Q_0(\beta_1)$.

Proposition 10.1. *Under the joint gauge, Einstein normal form, and closed-form transport assumptions above, a second-order compatible almost-Kähler deformation has $\beta_1 = 0$.*

Proof. Subtract the average from (38) to obtain $Q_0(\beta_1) = 0$. Equation (24) gives $\beta_1 = 0$. □

The conclusion concerns the independent first-order non-Kähler harmonic component. It does not exclude all metric deformations, or all higher-order emergence of nonintegrability, without further nonlinear analysis.

10.2 Uniformity along controlled ALE degeneration

Uniform length coercivity requires control beyond ordinary integral nonconcentration. One sufficient matching setting is the following: the curvature limits are Ricci-flat Kähler ALE spaces with groups in SU(2); on each neck the difference from the flat cone satisfies

$$r^k|\nabla^k e_j| \le \delta_j w_j(r), \qquad \int_{\text{neck}} w_j(r)\frac{dr}{r} \le C, \qquad \delta_j \to 0,$$

for the derivatives needed in the harmonic transfer equation; and the boundary traces of the forms match the orbifold and ALE limits. In the A_1 cone the homogeneous closed self-dual degrees are

$$\{\dots,-6,-4\} \ \cup \ \{0\} \ \cup \ \{2,4,\dots\}.$$

The zero mode is isolated. Integration of its perturbed equation changes it by at most $C\delta_j \int w_j(r)dr/r = o(1)$. Bounded harmonic coefficients on the complete Ricci-flat ALE limits are constant, while their L^2 cohomology is anti-self-dual. A blow-up argument at a maximum therefore yields

$$\|\psi\|_\infty \le C\|\psi\|_2, \qquad \mathcal{H}^+_{g_j} \longrightarrow \mathcal{H}^+_{g_\infty}.$$

Under these matching assumptions the compactness proof of (24) is uniform. The original conditional formulation and transfer estimates are recorded in [16], section "Coupled Kähler–Einstein deformations."

10.3 The normal period and a cscK comparison

A useful comparison separates a positive normal eigenvalue from the anti-Hermitian deformation kernel. For a cscK surface with scalar curvature $4E$, let

$$\rho_0 = \rho - E\omega.$$

It is closed, primitive, anti-self-dual, and harmonic. Thus

$$P(g) := \int |\operatorname{Ric}^\circ|^2 \, dV_g = 2\|\rho_0\|_2^2 = -2(2\pi c_1 - E[\omega])^2. \tag{39}$$

For a vanishing class $\delta^2 = -2$, $c_1 \cdot \delta = 0$, write $\zeta = \int_\delta \omega$. Then $\int_\delta \rho_0 = -E\zeta$. If $\theta = \delta^{1,1}$, $\theta^2 \to -2$, and the comparison classes take the normalized form

$$[\omega] = \lambda(2\pi c_1/E + \varepsilon\theta),$$

the cohomological calculation gives

$$\frac{P(g)}{E^2\zeta^2} = -\frac{2}{\theta^2} - \frac{E^2\varepsilon^2}{2\pi^2 c_1^2} \longrightarrow 1.$$

The corresponding Hermitian transverse-traceless mode satisfies $\Delta_E H_\theta = -2EH_\theta$. Its linearized Einstein eigenvalue is $-E > 0$, not a Kodaira–Spencer zero mode.

A local Einstein exclusion theorem would additionally require a uniform inverse on the complement, comparison families spanning all local period directions, an error $o(|\zeta|)$ in the relevant weighted norms, and coverage of all reduced parameters. Under those hypotheses the normal obstruction is $E\zeta + o(|\zeta|)$; degree then forces $\zeta = 0$, and the harmonic class $[\rho_0]$ vanishes. This describes a conditional completion of the local argument. The comparison identity (39) alone supplies none of those uniform analytic hypotheses.

11 Low-energy symplectic modifications and normalization

The following non-Kähler constructions complement the rigidity statements. The repository contains exact-collar low-energy metrics on an exceptional disk bundle for which

$$J_6(g) = \int |\operatorname{Ric}_g + 6g|^2 \, dV_g < 8\pi^2.$$

The initially transmitted closed self-dual form vanishes on the bolt. A compactly supported rotation of the orbit metric creates a nonzero exceptional period while preserving the outer collar. More precisely, if H_* is the orbit metric matrix and χ is one near the bolt and zero at the collar, use

$$H_\varepsilon = R(\varepsilon\chi) H_* R(\varepsilon\chi)^T.$$

Solving the closed self-dual transport equation from the outer Kähler form gives an exceptional period p satisfying

$$p'(0) = -\pi \int \chi'(r)\Phi(r)\, dr > 0, \qquad \Phi > 0.$$

Near the bolt the rotation is constant and extends smoothly; outside it is the identity. ODE uniqueness rules out a simultaneous zero on regular orbits, and the nonzero period removes the bolt zero. For sufficiently small ε the form is symplectic and $J_6 < 8\pi^2$ persists. The construction is given in [16], section "Symplectic closure of a low-cost block."

This gives a globally conformally almost-Kähler metric after insertion into the stated compact negative Kähler–Einstein backgrounds. It is not a construction of an Einstein metric. It also illustrates a significant normalization effect. Write

$$f_p = \frac{|\Omega_p|_{g_p}}{\sqrt{2}}, \qquad \widehat{g}_p = f_p g_p = v_p^2 g_p, \qquad v_p = \sqrt{f_p}.$$

If the normal bolt distance is r and its area is πb^2, the local expansion is

$$f_p^2 = a_0^2 p^2 + \kappa^2 r^2 + O(p^2 r^2 + r^4), \qquad a_0 = (\pi b^2)^{-1}.$$

With $D = \operatorname{div}\nabla = -\Delta$,

$$\frac{D v_p}{v_p} = \frac{\kappa^2(4a_0^2 p^2 + \kappa^2 r^2)}{4(a_0^2 p^2 + \kappa^2 r^2)^2} + O(1).$$

The scalar conformal formula and $dV = 2\pi^2 b^2 r\, dr + O(r^3) dr$ yield

$$\int s^2_{\widehat{g}_p} \, dV_{\widehat{g}_p} \sim \frac{C_E}{p^2}, \qquad C_E = 36(2\pi^2 b^2)\frac{7\kappa^2}{32 a_0^2}.$$

By conformal invariance of the Weyl integral,

$$P(\widehat{g}_p) - P(g_p) = \frac{1}{12}\left(\int s^2_{\widehat{g}_p} \, dV_{\widehat{g}_p} - \int s^2_{g_p} dV_{g_p}\right) \sim \frac{C_E}{12 p^2}.$$

Thus the smooth low-energy metrics and their constant-length almost-Kähler representatives have very different curvature energies as the exceptional period tends to zero. Conformal availability of a symplectic structure does not preserve an Einstein defect estimate.

12 The remaining closed problem

The preceding results impose simultaneous restrictions on a closed strict negative Einstein almost-Kähler structure: it has torsion zeros; the determinant potential is positive at every such zero; the action defect $D_W = \mathcal{T}$ is positive; it is not in a same-orientation Einstein component

containing a Kähler–Einstein metric or a noncollapsed Kähler–Einstein orbifold endpoint; and any available canonical currents satisfy (29)–(32). None of these restrictions is, by itself, a contradiction.

The differential obstruction still involves the transverse Weyl block and its coupling to the moving almost-Kähler line. Write $\nabla = D + C$ on $\mathbb{R}n \oplus n^{\perp}$, with

$$C_i = \begin{pmatrix} 0 & -p_i^T \\ p_i & 0 \end{pmatrix}, \qquad \sum_i p_i p_i^T = \frac{q}{4}\,\mathrm{Id}, \qquad D^* p = 2v.$$

The projected Einstein Weyl equation is

$$D^*DB + \left(\frac{3s}{2} + 2q\right) B = 2\sum_i [p_i (D_i v)^T + (D_i v) p_i^T]_0 + 2(vv^T)_0. \tag{40}$$

After integration by parts it gives

$$\int_M \left(|DB|^2 + \left(\frac{3s}{2} + 2q\right) |B|^2 + 4\sum_i \langle (D_i B) p_i, v\rangle - 10\langle Bv, v\rangle \right) dV_g = 0. \tag{41}$$

These equations are derived in [22]. Because $s < 0$, the zeroth-order coefficient is negative near torsion zeros. Neither discarding the mixed terms nor estimating them independently creates a positive identity. This is consistent with the complete noncompact example and with the permitted leading jets.

Question 12.1 (Closed negative Goldberg problem in dimension four)**.** Does every closed connected almost-Kähler four-manifold with $\mathrm{Ric} = -ag$, $a > 0$, satisfy $\nabla\Omega = 0$?

The integrability theorems above apply under the determinant-potential, curvature, canonical-orthogonality, or Einstein-component hypotheses. The complete noncompact example, linear compatibility conditions, and energy identities for noncollapsed degeneration describe complementary aspects of the problem. The general closed case remains open.

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